

Course: B. Tech.

Branch: Common to all branch

Semester: I

Subject Code & Name: 23UD1000BS101 & Engineering Mathematics-I

Max Marks: 60

Date: 13.01.2026

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	If $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 4 \\ 3 & 4 & 2 \end{bmatrix}$ then the eigen value of A are	CO1	1
	☒ a. (2, 2, 2) ☐ b. (2, 1, 3) ☐ c. (1, 2, 4) ☐ d. None		
2	A linear system is called consistent if it has	CO1	1
	☐ a. infinitely many solutions ☐ b. no solution ☒ c. at least one solution ☐ d. None		
3	If A and B are square matrices of the same order, then trace (AB) is equal to	CO1	1
	☐ a. trace (A + B) ☐ b. trace (A) trace (B) ☒ c. trace (BA) ☐ d. None		
4	If $u = (x^2 - 1)(y + 2)$ then the $\frac{\partial u}{\partial y}$ is	CO2	1
	☐ a. $2x(y + 2)$ ☐ b. $2y(x^2 - 1)$ ☐ c. $(x^2 - 1)(y + 2)$ ☒ d. None		
5	If $u = \log(x \tan^{-1} x)$ then $\frac{\partial u}{\partial x \partial y}$ is	CO2	1
	☒ a. 0 ☐ b. 1 ☐ c. 2 ☐ d. None		
6	If $x = r \cos \theta$, $y = r \sin \theta$ then $\frac{\partial(x, y)}{\partial(r, \theta)}$ is	CO2	1
	☒ a. r ☐ b. 2r ☐ c. 3r ☐ d. None		
7	If $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$ and $t = \frac{\partial^2 f}{\partial y^2}$, then the condition for saddle point is	CO3	1
	☒ a. $rt - s^2 < 0$ ☐ b. $rt - s^2 > 0$ ☐ c. $rt - s^2 = 0$ ☐ d. None		
8	The curve $y^2(a - x) = x^2(x + a)$ is symmetrical about	CO3	1
	☐ a. y axis ☒ b. x axis ☐ c. $y = x$ ☐ d. None		
9	The value of $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx$ is equal to	CO4	1
	☒ a. $\frac{3\pi}{8}$ ☐ b. $\frac{3\pi}{16}$ ☐ c. $\frac{3\pi}{25}$ ☐ d. None		
10	The number of loops in polar curve $r = a \cos 4\theta$	CO4	1
	☒ a. 8 ☐ b. 4 ☐ c. 2 ☐ d. None		

11	The value of $\int_0^1 \int_0^x dx dy$ is equal to	CO5	1
	a. $\frac{1}{2}$ b. $\frac{1}{3}$ c. $\frac{1}{4}$ d. None		
12	The value of $\int_0^1 \int_0^x \int_0^{xy} x dx dy dz$ is equal to	CO5	1
	a. $\frac{1}{5}$ b. $\frac{1}{7}$ c. $\frac{1}{10}$ d. None		
Q.2	Solve the following.		12
A)	Find the rank of a matrix A by reducing it to normal form, where $A = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$.	CO1	6✓
B)	Find the eigen values and the eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.	CO1	$\frac{6}{3}$
Q.3	Solve the following.		12
A)	If $z(x+y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.	CO2	6 ✓
B)	If $u = \operatorname{cosec}^{-1} \left(\frac{1}{x^2+y^2} \right)^{\frac{1}{2}}$, Prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{12} \left[\frac{13}{12} + \frac{\tan^2 u}{12} \right]$.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	If $u = x + y + z$, $uv = y + z$ and $uvw = z$, show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2 v$.	CO3	6
B)	Expand $f(x,y) = e^{xy}$ at $(1,1)$.	CO3	6
C)	Divide 24 into three parts such that the continued product of the first, square of the second and cube of the third may be maximum.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Evaluate $\int_0^\infty \frac{x^6}{(1+x^2)^7} dx$.	CO4	6
B)	Trace the curve $y^2 = \frac{x^2(a^2-x^2)}{a^2+x^2}$.	CO4	6
C)	Trace the curve $r = a(1 + \cos \theta)$.	CO4	6

Q.6	Solve Any Two of the following.		12
A)	Evaluate $\int_0^1 \int_{x^3}^{\sqrt{x}} (4xy - y^3) dx dy$.	CO5	6
B)	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dz dy dx$.	CO5	6
C)	Find the area between $r = 2 \sin \theta$ and $r = 4 \sin \theta$.	CO5	6

2

1